COMPUTER

Concise study notes for the
OAVS Computer Teacher Examination
Based strictly on the syllabus topics listed below

Syllabus (as given)
Computer : Block Diagram of elements of digital computer-their functions. Memory, CPU, I-O devices, Secondary storages, Magnetic Tape, Disk, CD-ROM. Other recent developments-Scanners, Digitizer, Plotters. Hardware and Software. Micro, Mini and Main-frame computers-their features.

Contents: 1 Block Diagram of Elements of Digital Computer | 2 Memory | 3 CPU | 4 I-O Devices | 5 Secondary Storages | 6 Magnetic Tape, Disk, CD-ROM | 7 Other Recent Developments: Scanners, Digitizer, Plotters | 8 Hardware and Software | 9 Micro, Mini and Main-frame Computers | Quick Revision.

EXAM POINT	Yellow boxes = likely exam points.
DON'T CONFUSE	Red boxes = commonly confused pairs.
REMEMBER	Green boxes = key facts to memorise.

1. Block Diagram of Elements of Digital Computer—Their Functions

A **digital computer** is an electronic machine that accepts data and instructions in discrete (binary: 0 and 1) form, stores them, processes them under the control of a stored program, and produces results (information). It works on the **stored-program concept** (John von Neumann): the program and the data are kept together in the same memory. (An analog computer, in contrast, measures continuous quantities such as voltage or temperature.)

Every digital computer is organised into **five functional units**: Input unit, Memory (storage) unit, Arithmetic Logic Unit, Control Unit and Output unit. The ALU and the Control Unit together form the **CPU**.

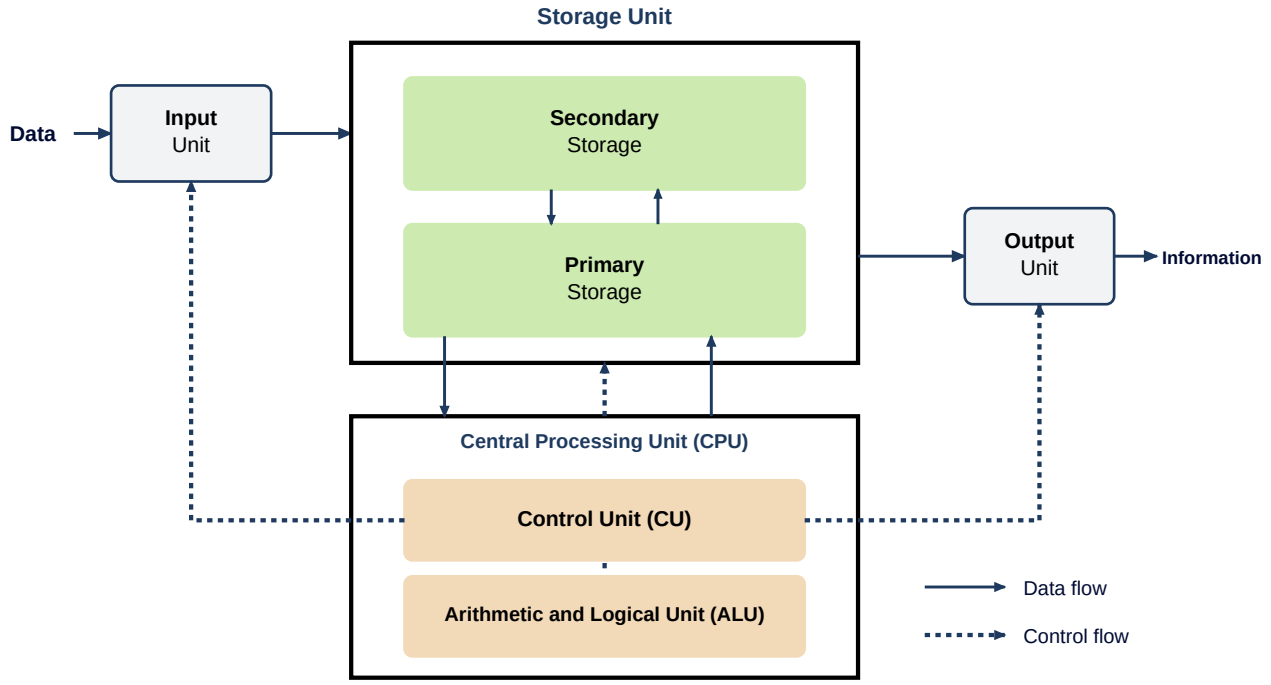


Fig. 1: Block diagram of a digital computer (data flow and control flow)

Unit	Function	Examples
Input unit	Accepts data and programs from the user, converts them to binary and passes them to the computer.	Keyboard, mouse, scanner
Memory (storage) unit	Stores the program, input data, intermediate results and final results. Main memory is directly accessible by the CPU; secondary storage keeps data permanently.	RAM, ROM, hard disk
Arithmetic Logic Unit (ALU)	Performs arithmetic operations (+, -, x, /) and logical operations (AND, OR, NOT, comparisons).	Part of CPU
Control Unit (CU)	Fetches and decodes instructions, directs and coordinates all other units by control signals. It does not process data itself.	Part of CPU
Output unit	Converts binary results into a form humans can understand and delivers them.	Monitor, printer, plotter

Flow of data between the components

- **Input:** program and data enter through an input device and are converted to binary.
- **Storage:** they are placed in main memory (RAM); permanent copies are kept in secondary storage.
- **Control:** the CU fetches an instruction from memory, decodes it and sends control signals.
- **Processing:** the ALU operates on data taken from memory / registers; results go back to memory.
- **Output:** final results are sent from memory to the output unit as information.

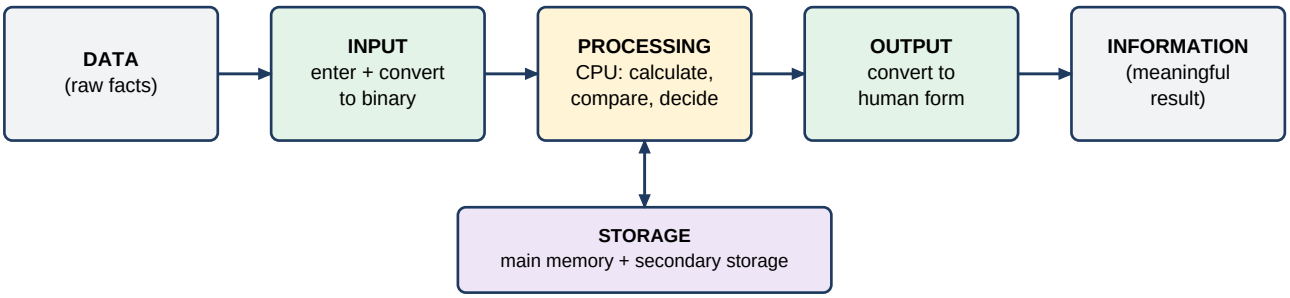


Fig. 2: Input-Processing-Output (IPO) flow with storage

EXAM POINT Five units: Input, Memory, ALU, CU, Output. **CPU = ALU + CU** (registers are inside the CPU). Data = raw facts; Information = processed, meaningful data. Control Unit = 'manager' of the computer.

2. Memory

Memory is the computer's storage area where programs, data and results are held so that the CPU can use them.
Functions: (i) hold the program being executed, (ii) hold input data, (iii) hold intermediate and final results, (iv) supply instructions and data to the CPU, and receive results from it.

Units of memory

Unit	Value	Unit	Value
Bit	0 or 1 (smallest unit)	Megabyte (MB)	1024 KB
Nibble	4 bits	Gigabyte (GB)	1024 MB
Byte	8 bits (one character)	Terabyte (TB)	1024 GB
Kilobyte (KB)	1024 bytes	Word	Group of bits handled together by the CPU (e.g. 32 / 64 bits)

Classification of memory

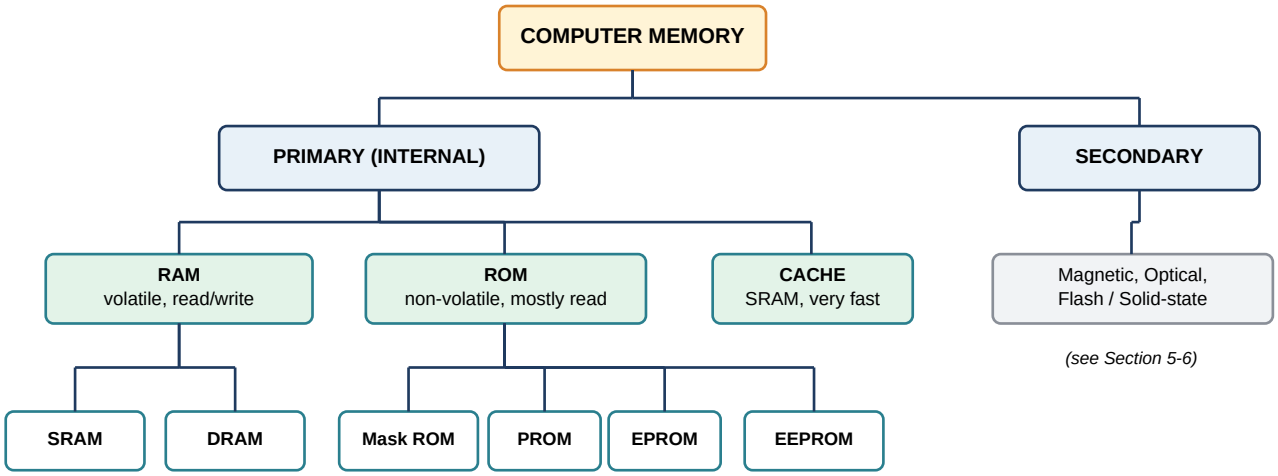


Fig. 3: Classification of computer memory

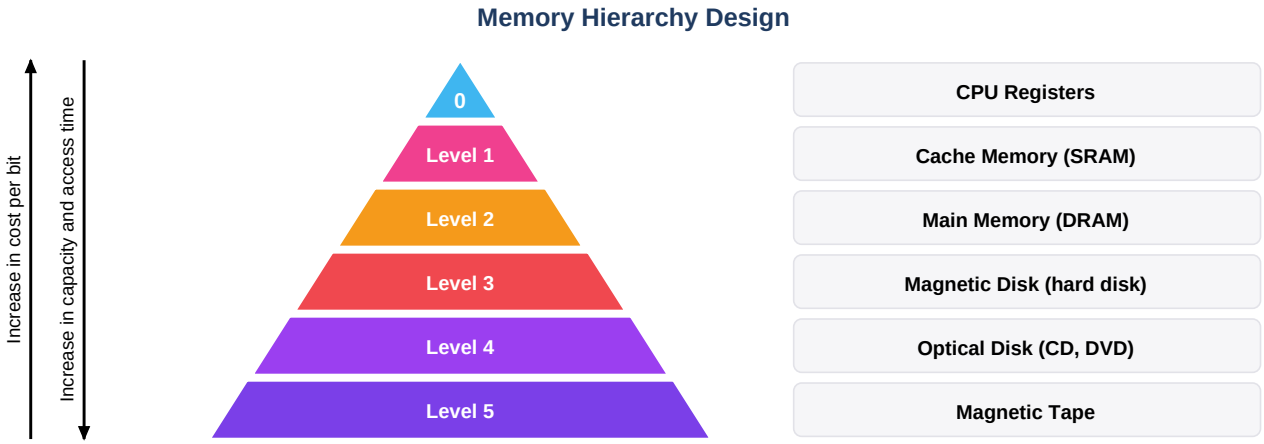
Types, characteristics and uses

Type	Key characteristics	Typical use
Registers	Inside the CPU; smallest and fastest; very costly; few bytes each.	Hold current instruction, address, operand
Cache (SRAM)	Small, very fast memory between CPU and RAM; stores frequently used data. Levels L1 (inside CPU, fastest), L2, L3. *Cache hit* = data found; *miss* = fetched from RAM.	Reduces CPU waiting time
RAM (Random Access Memory)	Read/write, **volatile** (contents lost when power is off); any location is reached in equal time.	Main working memory
SRAM (Static RAM)	Uses flip-flops; no refreshing; faster, costlier, low density.	Cache
DRAM (Dynamic RAM)	Uses capacitor + transistor; needs periodic **refreshing**; slower, cheaper, high density.	Main memory
ROM (Read Only Memory)	**Non-volatile**; contents are fixed or rarely changed; holds firmware.	Boot program (BIOS)
Mask ROM	Programmed by the manufacturer; cannot be changed.	Mass-produced fixed programs
PROM (Programmable ROM)	Programmable once by the user with a special device.	Custom firmware
EPROM (Erasable PROM)	Erased by **ultraviolet** light, then reprogrammed.	Development / testing
EEPROM (Electrically Erasable PROM) / Flash	Erased **electrically** (Flash erases in blocks); non-volatile.	BIOS update, pen drives, SSD

Feature	RAM	ROM
Volatility	Volatile	Non-volatile
Operations	Read and write	Read only (reprogramming needs special process)

Feature	RAM	ROM
Purpose	Temporary storage of running programs and data	Permanent storage of start-up (boot) instructions
Speed / cost	Fast; costlier per byte than disk	Slower to change; cheap in bulk

Memory organisation



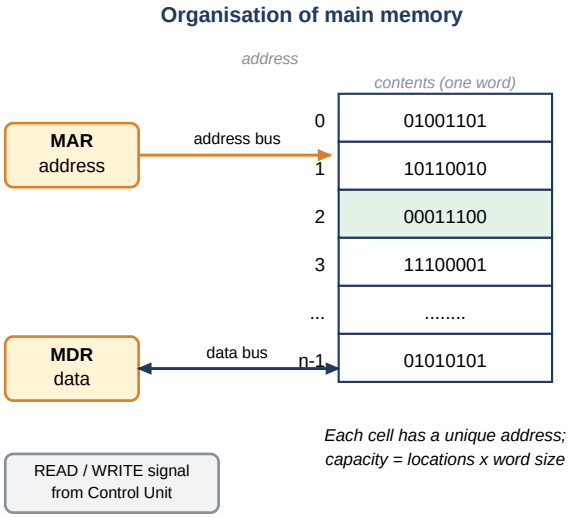


Fig. 4: Memory hierarchy and organisation of main memory

- Main memory is an array of **cells**; each cell has a unique **address** and stores one word / byte.
- **MAR** (Memory Address Register) holds the address; **MDR** (Memory Data Register) holds the data being read or written; the CU (Control Unit) gives the read/write signal.
- **Read** is non-destructive (the cell keeps its contents); **write** is destructive (old contents are overwritten).

Numerical: Number of locations = 2^n (n = address lines); Capacity = number of locations x word size.

Question	Solution
Q1. A memory has 12 address lines and a word size of 8 bits. Find the number of locations and the capacity.	Locations = 2^{12} = 4096 (4 K). Capacity = 4096 x 8 bits = 32,768 bits = 4096 bytes = **4 KB**.
Q2. How many address lines are needed for a 64 KB memory with 1 byte per location?	64 KB = 64 x 1024 = 65,536 = 2^{16} locations. Address lines needed = **16**.

DON'T CONFUSE **Cache vs RAM:** cache is smaller, faster, SRAM, placed near the CPU. **RAM vs ROM:** volatile vs non-volatile. **Primary vs secondary:** primary is directly accessed by the CPU. **Bit vs byte:** 8 bits = 1 byte.

3. CPU

The **Central Processing Unit (CPU)** is the 'brain' of the computer. It interprets and executes instructions, performs calculations and decisions, and controls all other parts. In microcomputers the entire CPU is built on a single chip called a **microprocessor** (first: Intel 4004, 1971). Its major components are the **Control Unit**, the **ALU** and **Registers**.

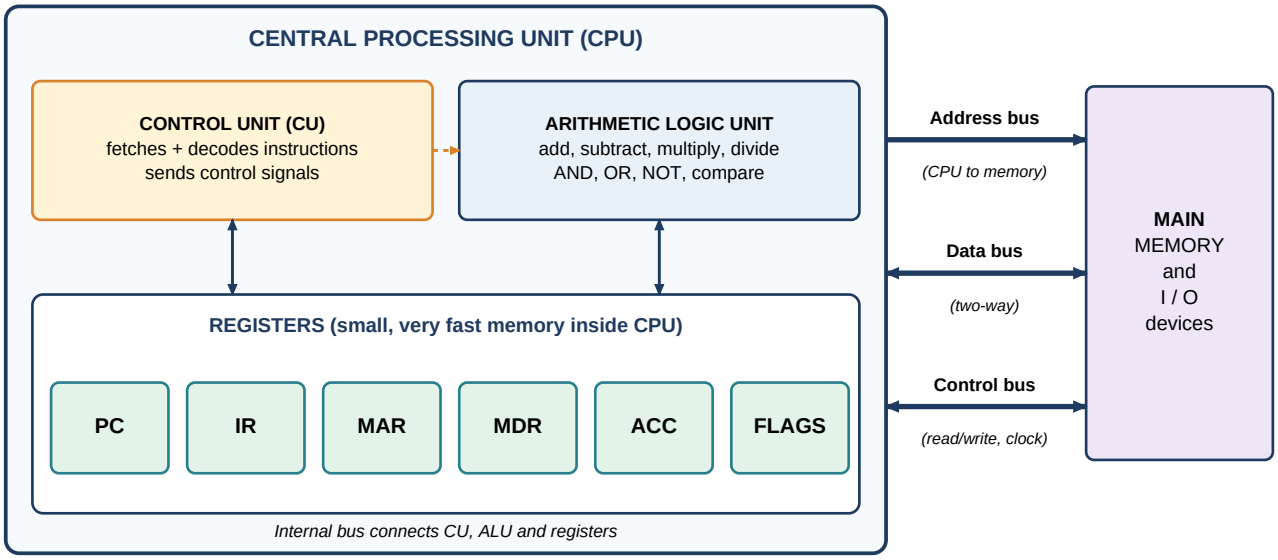


Fig. 5: CPU and its major components, with the three system buses

Component	Role
Control Unit (CU)	Fetches instructions, decodes them, generates timing and control signals for ALU, memory and I/O; controls sequence of execution.
ALU	Arithmetic (add, subtract, multiply, divide) and logic (AND, OR, NOT, comparison: <, =, >) operations.
PC (Program Counter)	Holds the address of the **next** instruction to be fetched.
IR (Instruction Register)	Holds the instruction currently being decoded / executed.
MAR (Memory Address Register) / MDR (Memory Data Register, also MBR)	MAR: address of the memory location to access. MDR: data just read from, or about to be written to, memory.
ACC (Accumulator)	Holds intermediate results of ALU operations.
Flags / Status register	Bits showing results of operations (zero, carry, overflow, sign).

Buses: *address bus* (one-way, CPU to memory/I-O, carries addresses), *data bus* (two-way, carries data), *control bus* (carries read/write, clock and other control signals). **CPU performance** depends on clock speed (MHz/GHz), word size (bits), cache size, and number of cores.

Basic instruction-processing flow (instruction cycle)

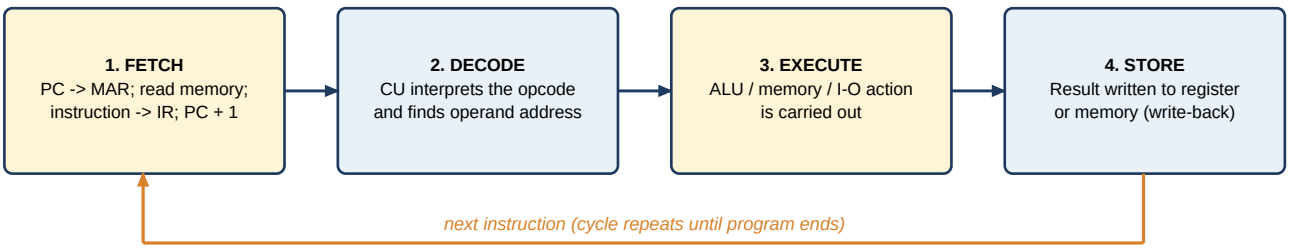


Fig. 6: Fetch-Decode-Execute-Store cycle

EXAM POINT Fetch: address in PC is copied to MAR, instruction is read into MDR then IR, and PC is incremented. The CU decodes, the ALU executes. **CPU = CU + ALU + Registers**; the CU controls but does not compute.

DON'T CONFUSE **CPU vs microprocessor:** a microprocessor is a CPU on one chip. **CU vs ALU:** CU controls, ALU computes. **PC vs IR:** PC = next instruction address; IR = current instruction.

4. I-O Devices

Input devices feed data and instructions into the computer and convert them into binary. **Output devices** convert the computer's binary results into a human-readable form. Together they are the *peripheral* devices that connect the user to the CPU.

Classification and examples

Category	Devices	Function / note
Keying device	Keyboard	Types characters, numbers and commands
Pointing devices	Mouse, trackball, joystick, light pen, touch screen	Move a cursor and select objects
Scanning / source-data devices	Scanner, barcode reader (product codes), OCR (printed characters), **OMR** (marks, e.g. answer sheets), **MICR** (magnetic ink characters on bank cheques)	Read data directly from documents
Audio / video input	Microphone, webcam, digital camera	Capture sound, images, video
Graphics input	Digitizer / graphics tablet	Capture drawing coordinates

Category	Devices	Function / note
Soft-copy output	Monitor (CRT, LCD, LED), speakers, projector	Temporary display; no paper copy
Hard-copy output	Printers, plotters	Permanent copy on paper
Impact printers	Dot-matrix, line, daisy-wheel	Strike the paper through a ribbon; noisy; can make carbon copies
Non-impact printers	Inkjet, laser, thermal	Quiet, faster, better quality; no carbon copies
Both input and output	Touch screen; storage devices (disk, pen drive)	Send data to and receive data from the computer

Basis	Input devices	Output devices
Direction of data	From user to computer	From computer to user
Conversion	Human-readable to binary	Binary to human-readable
Examples	Keyboard, mouse, scanner, microphone	Monitor, printer, plotter, speaker
Performance measures	Resolution, scan speed (scanner); typing rate	Monitor: resolution (pixels), refresh rate. Printer: DPI, speed in CPS / LPM / PPM

EXAM POINT OMR = answer sheets; MICR = bank cheques; OCR = printed text to editable text; barcode reader = shops. Impact printers (dot-matrix) are the only ones that can print carbon copies.

5. Secondary Storages

Secondary storage (auxiliary or backing store) is permanent, **non-volatile** storage that keeps programs and data when the power is off. The CPU cannot execute from it directly; data must first be copied into main memory.

- **Purpose:** long-term storage, backup and archiving, transfer of data between computers.
- **Characteristics:** very large capacity, low cost per bit, non-volatile, slower than primary memory, often removable.
- **Types:** magnetic (hard disk, floppy disk, magnetic tape), optical (CD, DVD), solid-state / flash (SSD, pen drive, memory card).
- **Access methods:** *sequential* (records read one after another: tape) and *direct / random* (go straight to a location: disk, CD).

Basis	Primary memory	Secondary storage
Access by CPU	Direct	Indirect, through main memory
Speed	Very fast (nanoseconds)	Slower (milliseconds)
Capacity	Smaller	Much larger
Cost per byte	High	Low
Volatility	RAM volatile; ROM non-volatile	Non-volatile
Examples	RAM, ROM, cache	Hard disk, tape, CD-ROM, pen drive

EXAM POINT Secondary storage is non-volatile, slower and cheaper than primary memory; the CPU can't execute programs directly from it.

6. Magnetic Tape, Disk, CD-ROM

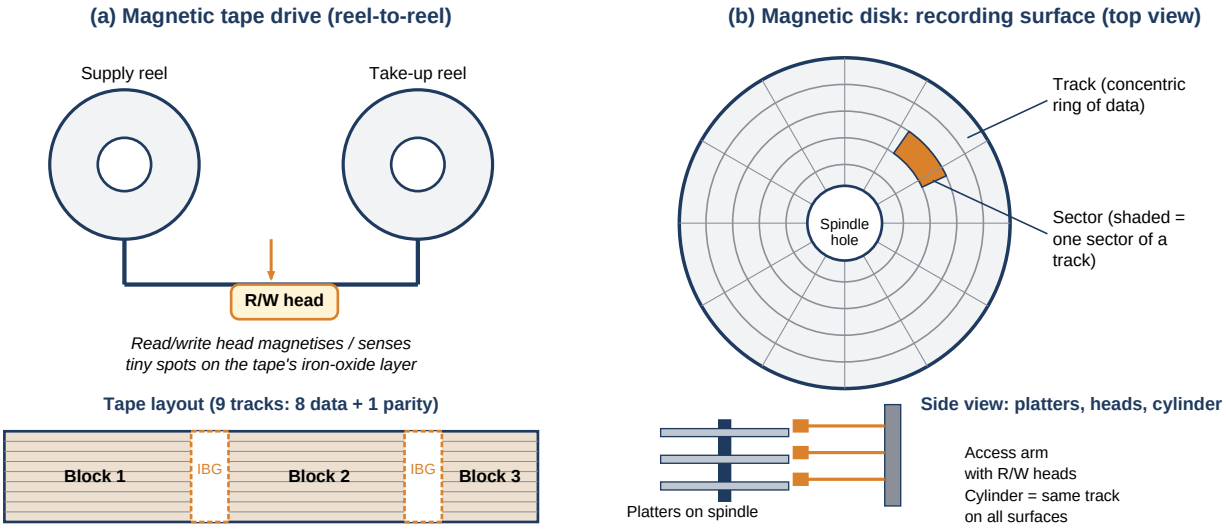


Fig. 7: Magnetic tape (drive and layout) and magnetic disk (surface and side view)

Magnetic tape

A plastic (polyester) tape coated with iron oxide, wound on reels or in cartridges. Data is recorded as magnetised spots across **tracks** (9 tracks: 8 data bits + 1 parity bit on classic tapes). Records are grouped into **blocks** separated by **inter-block gaps (IBG)** that let the tape start and stop; *BOT / EOT* markers show beginning and end of tape. Recording density is measured in bits per inch (bpi).

- **Working:** the tape moves past the read/write head; magnetising spots writes data, sensing them reads data. Access is **sequential**.
- **Advantages:** very cheap per bit, huge capacity (modern cartridges hold terabytes), portable, good for backup and archives.
- **Limitations:** slow to reach a record (must wind through earlier data), no direct access, needs a clean and cool store.
- **Uses:** backup, archiving, batch processing (e.g. payroll) on mainframes.

Magnetic disk

Circular platters (aluminium or glass) coated with magnetic material, spinning on a **spindle** (e.g. 5400 or 7200 rpm). Each surface has concentric **tracks** divided into **sectors** (smallest addressable unit, typically 512 bytes). A **cylinder** is the set of same-numbered tracks on all surfaces. Read/write heads on an access arm move across the surface. Types: floppy disk (3.5 inch, 1.44 MB; obsolete) and hard disk (fixed, hundreds of GB to several TB).

- **Disk capacity** = surfaces x tracks per surface x sectors per track x bytes per sector. Example: 8 x 1000 x 100 x 512 = 409,600,000 bytes (about 410 MB).
- **Access time** = seek time (move head to track) + rotational latency (wait for sector) + transfer time.
- **Advantages:** direct (random) access, fast, large capacity, rewritable. **Limitations:** moving parts can fail (head crash), sensitive to shock, dust and magnets, costlier than tape per bit.
- **Uses:** storing the system, programs and files; databases; online storage.

CD-ROM (Compact Disc Read-Only Memory)

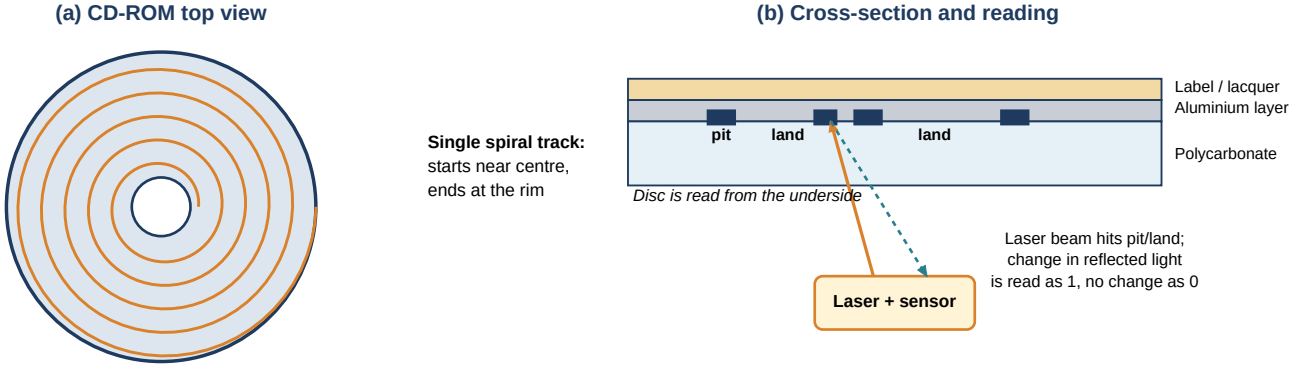


Fig. 8: CD-ROM spiral track and the pit / land reading principle

A 120 mm polycarbonate disc coated with a reflective aluminium layer and a protective lacquer. Data is stamped by the manufacturer as microscopic **pits** (depressions) and **lands** (flat areas) along a **single spiral track** from the centre outward. A low-power **infrared laser** (about 780 nm) reads it: a change between pit and land is read as 1, no change as 0.

- **Capacity:** about 650-700 MB. **Speed:** 1x = 150 KB/s; drives are rated 24x, 52x, etc.
- **Advantages:** cheap to mass-produce, portable, durable, good for distribution of software and multimedia.
- **Limitations:** read-only (data cannot be changed), slower than hard disk, scratches damage data.
- **Related:** CD-R (record once), CD-RW (rewritable). **Uses:** software, encyclopaedias, audio and video, educational content.

Basis	Magnetic tape	Magnetic disk	CD-ROM
Medium	Iron-oxide coated plastic tape	Magnetic-coated platters	Polycarbonate disc, aluminium layer
Technology	Magnetic	Magnetic	Optical (laser)
Storage layout	Tracks and blocks along the tape	Concentric tracks, sectors, cylinders	Single spiral track of pits and lands
Access method	**Sequential**	**Direct (random)**	Direct (slower seek than hard disk)
Capacity (typical)	Reels: tens to hundreds of MB; cartridges: TB	Floppy 1.44 MB; hard disk GB to TB	650-700 MB
Speed	Slowest	Fast	Moderate
Rewritable	Yes	Yes	No (CD-ROM); CD-RW yes
Best use	Backup, archive	Online working storage	Software / multimedia distribution

DON'T CONFUSE **Sequential vs direct access:** tape = sequential, disk and CD-ROM = direct. **CD-ROM vs CD-R vs CD-RW:** read-only, write-once, rewritable. **Seek time vs rotational latency:** head movement vs waiting for the sector to come under the head.

7. Other Recent Developments: Scanners, Digitizer, Plotters

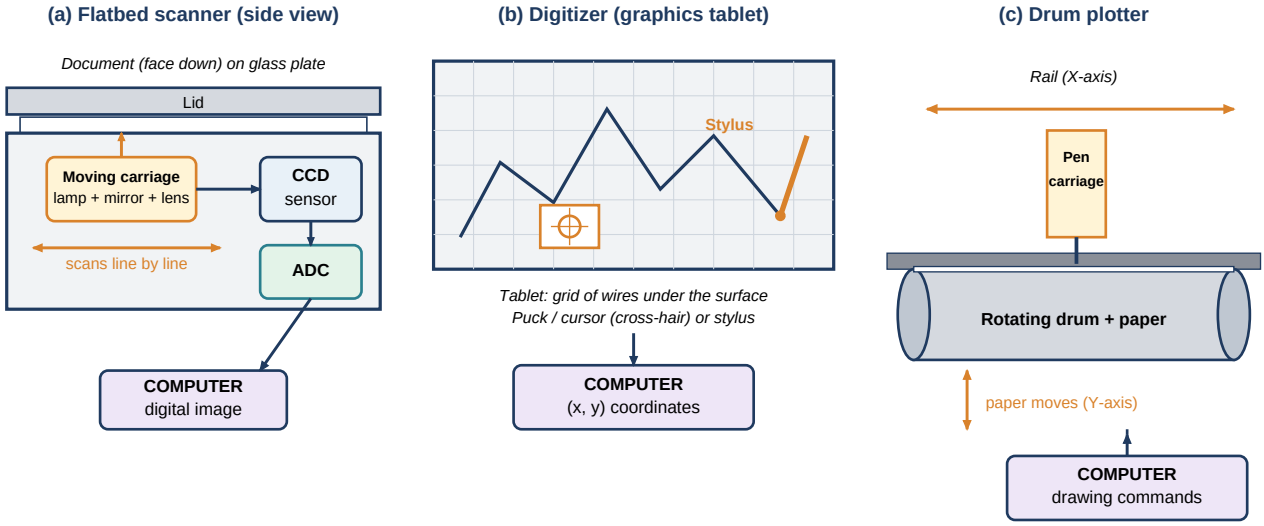


Fig. 9: Scanner, digitizer and plotter (schematic)

Scanner (input device)

A **scanner** converts a paper document or photograph into a digital image. A lamp lights the document; reflected light passes through mirrors and a lens to a **CCD** (charge-coupled device) or CIS sensor, which produces electrical signals; an **ADC** turns them into digital pixels. Quality is measured in **dpi** (dots per inch). With **OCR** software a scanned text image can be converted to editable text.

- **Types:** flatbed, sheet-fed, handheld, drum (highest quality; printing industry), film / slide scanner; barcode scanner for product codes.
- **Applications:** digitising records and books, photos, ID cards and signatures, office document archiving.

Digitizer (input device)

A **digitizer** (digitizing or graphics tablet) converts hand-drawn lines and positions into digital **x-y coordinates**. It has a flat tablet with a grid of fine wires beneath its surface and a **stylus** (pen) or **puck / cursor** (with cross-hair). Electromagnetic signals between grid and pointer give the pointer's exact position. Resolution is measured in lines per inch (lpi).

- **Applications:** CAD / architecture and engineering drawings, tracing maps (GIS), animation, and digital art.
- A digitizer captures **vector coordinates** of what you trace; it does not scan the whole picture.

Plotters (output device)

A **plotter** draws high-quality, large-size graphics such as drawings, maps and charts by moving pens (or an inkjet head) under computer control with **vector** commands, making continuous lines rather than dots.

Type	Working principle
Drum plotter	Paper moves back and forth over a rotating drum (Y-axis) while the pen carriage moves along a rail (X-axis).
Flatbed plotter	Paper lies fixed on a flat bed; the pen moves in both X and Y directions on a gantry.
Inkjet (wide-format) plotter	Modern type; sprays ink on wide paper rolls for posters, maps and CAD prints.
Cutting plotter	Uses a knife instead of a pen to cut vinyl or paper shapes.

- **Applications:** CAD, architecture, civil and mechanical drawings, maps, posters, signboards.
- **Advantages:** very accurate, handles large sheets. **Limitations:** slower and costlier than ordinary printers.

Basis	Scanner	Digitizer	Plotter
Device type	Input	Input	Output
Function	Converts a document / photo to a digital image (raster pixels)	Converts traced drawing to x-y coordinates (vector)	Draws vector graphics on paper
Typical measure	dpi	lpi / accuracy	Pen speed, paper size
Used in	Document archiving, photo digitising	CAD, map tracing	CAD / engineering drawings, maps

DON'T CONFUSE **Scanner vs digitizer:** scanner copies the whole image as pixels; digitizer records coordinates of what you trace. **Plotter vs printer:** plotter uses pens for continuous vector lines on large sheets; printers form characters from dots.

8. Hardware and Software

Hardware is the physical, tangible parts of a computer that you can touch: CPU, memory, I/O devices, storage devices and cables. **Software** is the set of programs and instructions that tells the hardware what to do; it cannot be touched. Hardware cannot work without software, and software cannot run without hardware: **software supplies the instructions, hardware does the work**. **Firmware** (e.g. BIOS) is software permanently stored in ROM chips.

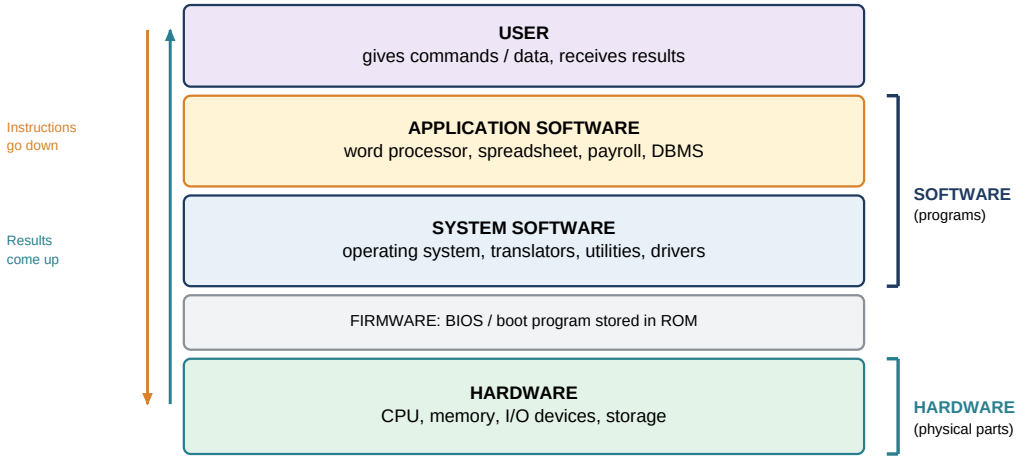


Fig. 10: Hardware-software relationship (layers)

Classification of software

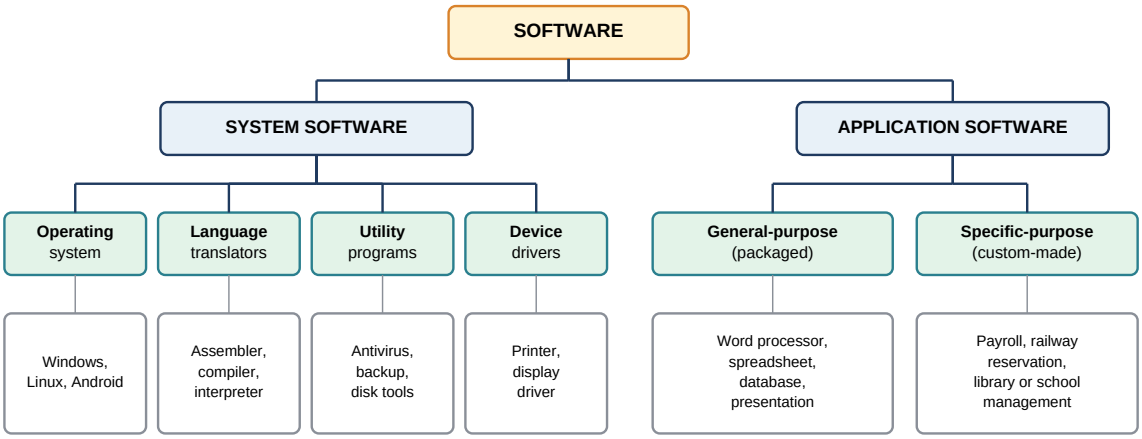


Fig. 11: Classification of software with examples

- **System software** manages and supports the hardware and runs application software. It includes the operating system, language translators (*assembler*: assembly to machine code; *compiler*: whole program at once; *interpreter*: line by line), utility programs and device drivers.
- **Application software** performs tasks for the user. *General-purpose / packaged* software suits many users (word processor, spreadsheet, database, presentation). *Specific-purpose / custom* software is written for one need (payroll, reservation).
- **By licence:** proprietary (commercial), open-source (source code freely available), freeware (free to use), shareware (free trial, then pay).

Basis	Hardware	Software
Nature	Physical, tangible	Logical, intangible (programs)
Examples	CPU, monitor, keyboard, RAM, disk	Word processor, compiler, utilities
Damage	Wears out, physically damaged	Does not wear out; corrupted by bugs or viruses
Dependence	Needs software to function	Needs hardware to run

EXAM POINT Firmware = software in ROM (e.g. BIOS). System software runs the computer; application software serves the user. Compiler translates the whole program; interpreter translates line by line.

9. Micro, Mini and Main-frame Computers—Their Features

Computers are classified by size, speed, memory and number of users. The three classes in the syllabus are described below.

Class	Features	Processing capability	Applications / typical users
Microcomputer	Built around a single **microprocessor**; small, cheap, low power; generally single-user. Forms: desktop, laptop, tablet.	Typically 32 / 64-bit word; GHz clock; GBs of memory.	Home, school, office, small business; individuals and students.
Minicomputer	Mid-sized, **multi-user** with time-sharing; first appeared in mid-1960s (e.g. PDP-8, PDP-11, VAX); modern 'midrange servers' continue this role.	Faster and larger memory than micro; supports tens to a few hundred terminals.	Small and medium organisations, departments, laboratories, manufacturing and process control.
Mainframe	Large, powerful, **centralised**; supports many terminals; very high reliability and security; needs an air-conditioned room; very expensive.	Very high speed; huge memory and storage; many I/O channels; multiprocessing; processes millions of transactions.	Banks, airlines, railways, insurance, government, census, large corporations.

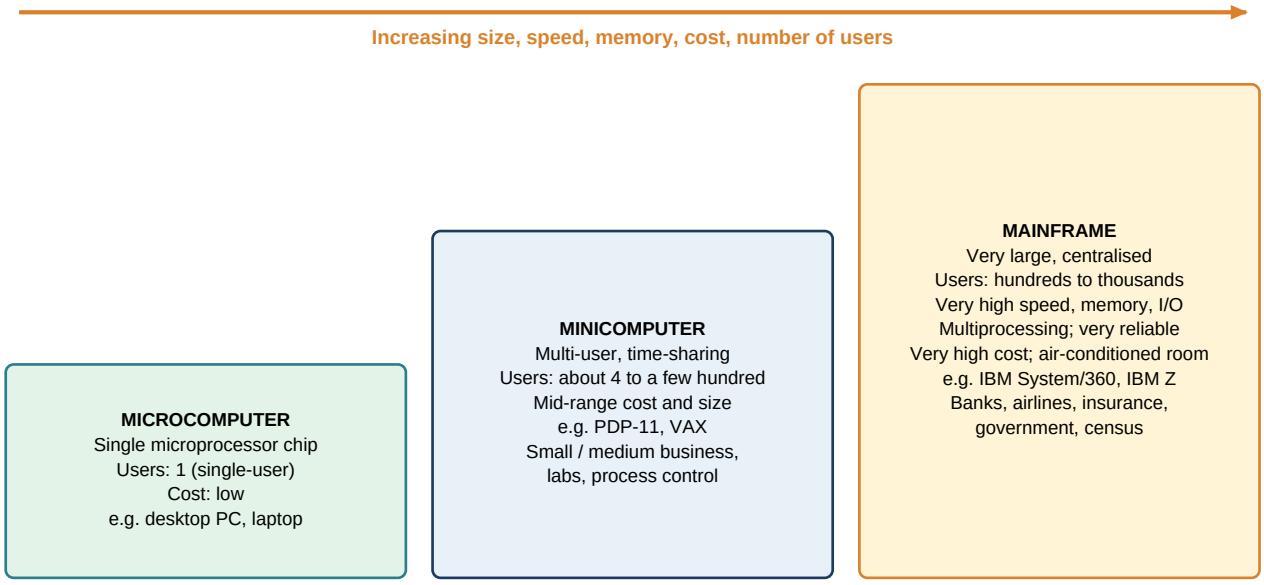


Fig. 12: Comparison of microcomputer, minicomputer and mainframe

Basis	Microcomputer	Minicomputer	Mainframe
Size	Smallest (desk / lap / hand)	Medium (cabinet)	Very large (room-sized)
Users	One (single-user)	Multi-user (about 4 to a few hundred)	Hundreds to thousands
Speed and memory	Lowest	Medium	Highest
Cost	Low	Moderate	Very high
Reliability / environment	Ordinary room	Needs moderate care	Strict air-conditioned environment
Examples	IBM PC, laptop, Macintosh	PDP-11, VAX	IBM System/360, IBM Z series

EXAM POINT Micro = single-user, microprocessor-based; Mini = multi-user, mid-range; Mainframe = very large, thousands of users, banks / airlines. Remember the order of increasing power: **Micro < Mini < Mainframe**.

QUICK REVISION

A. Key definitions

Term	Definition
Digital computer	Electronic machine that processes binary data under the control of a stored program.
CPU	Brain of the computer: Control Unit + ALU + registers.
Control Unit / ALU	Directs operations of all units / performs arithmetic and logical operations.
Register	Small, fastest storage inside the CPU.
Memory	Storage area holding programs, data and results for the CPU.
RAM / ROM	Volatile read-write main memory / non-volatile read-only memory.
Cache	Small, fast SRAM between CPU and RAM storing frequently used data.
Input / output device	Brings data into the computer / delivers results to the user.
Secondary storage	Permanent, non-volatile, large-capacity storage (disk, tape, CD, flash).
Magnetic tape	Sequential-access magnetic storage on a coated plastic tape; used for backup.
Magnetic disk	Direct-access magnetic storage on tracks and sectors of rotating platters.
CD-ROM	Optical read-only disc storing 650-700 MB as pits and lands on a spiral track.
Scanner / Digitizer / Plotter	Converts documents to images / converts drawings to x-y coordinates / draws large vector graphics.
Hardware / Software	Physical parts / programs that direct the hardware.
Firmware	Software stored permanently in ROM.
Micro / Mini / Mainframe	Single-user PC / multi-user mid-range / very large high-capacity centralised computer.

B. Memory and storage at a glance

Medium	Speed	Cost per bit	Capacity	Volatile?	Access
Registers	Fastest	Highest	Smallest	Yes	Direct
Cache (SRAM)	Very fast	Very high	KB to MB	Yes	Direct
RAM (DRAM)	Fast	High	GB	Yes	Random
Hard disk	Moderate	Low	Up to TB	No	Direct
CD-ROM	Moderate / slow	Very low	650-700 MB	No	Direct
Magnetic tape	Slowest	Lowest	Very large	No	Sequential

C. Important facts

- Five functional units: Input, Memory, ALU, CU, Output; **CPU = ALU + CU**.
- Instruction cycle: **Fetch, Decode, Execute, Store**; PC holds next instruction address, IR holds current instruction.
- Hierarchy (fast to slow, small to large): Registers, Cache, Main memory, Secondary, Offline storage.
- SRAM = cache (no refresh); DRAM = main memory (refreshed). EPROM = UV erase; EEPROM / Flash = electrical erase; PROM = programmable once.
- 1 byte = 8 bits; 1 KB = 1024 bytes; n address lines give 2^n locations.
- Tape = sequential; disk and CD-ROM = direct. Disk access time = seek + rotational latency + transfer time.
- Disk capacity = surfaces x tracks x sectors x bytes per sector. CD-ROM uses a laser to read pits and lands on a single spiral track.
- OMR = answer sheets, MICR = bank cheques, OCR = printed text, barcode reader = product codes.
- Impact printers (dot matrix) make carbon copies; non-impact (inkjet, laser) are quiet and faster.
- Scanner = input, raster image (dpi); digitizer = input, x-y coordinates; plotter = output, vector drawings.
- Hardware is physical; software is programs; firmware is software in ROM. System software runs the machine; application software serves the user.
- Micro = single-user; Mini = multi-user mid-range; Mainframe = largest, banks / airlines / government.

D. Likely examination questions

Multiple-choice (answers in brackets)

No.	Question	Answer
1	Which unit directs the operations of all other units?	Control Unit
2	Which register holds the address of the next instruction?	Program Counter (PC)
3	Which memory is volatile?	RAM
4	Which memory is erased with ultraviolet light?	EPROM
5	Cache memory is made of which type of RAM?	SRAM
6	Which storage medium uses sequential access?	Magnetic tape
7	Which of the following is read with a laser?	CD-ROM
8	Which device converts a paper document into a digital image?	Scanner
9	Which input device captures x-y coordinates for CAD work?	Digitizer
10	Which output device draws large engineering drawings?	Plotter
11	BIOS is an example of what?	Firmware (stored in ROM)
12	Which computer class serves thousands of users in banks and airlines?	Mainframe
13	MICR is used mainly in which field?	Banking (cheque processing)
14	Disk access time is the sum of which times?	Seek + rotational latency + transfer time

Short / descriptive questions

- Draw and label the block diagram of a digital computer and explain the function of each unit.
- Draw a labelled diagram of the CPU and explain the instruction cycle.
- Differentiate between RAM and ROM; SRAM and DRAM; primary memory and secondary storage.
- Name the types of ROM and state how each is programmed / erased.
- Explain the working of magnetic tape and magnetic disk. Compare sequential and direct access.
- Describe how data is stored and read on a CD-ROM.
- Explain scanner, digitizer and plotter with their uses. Differentiate scanner from digitizer.
- Define hardware and software. Classify software with examples.
- Compare microcomputers, minicomputers and mainframe computers in a table.
- Write short notes on: cache memory; impact and non-impact printers; input and output devices; firmware.